Project Objective. The goal of this assignment is to design and develop a functional website or software prototype that adheres to Human-Computer Interaction (HCI) standards. You must be prepared to present your work and defend your design decisions based on usability principles.

The object of analysis. The user interface of your website or software

Technical Requirements

- **Technology Stack.** You are free to choose any programming language, framework, or CMS (e.g., HTML/CSS/JS, React, Python, WordPress, etc.).
- **Structure.** The product must include a functional navigation system. It must have a home page (start window) and at least two levels of navigation depth (sub-menus or secondary pages).
- **Functionality.** While the backend logic can be minimal, the user interface (UI) must be fully interactive.

Core HCI Evaluation Criteria. During the development and presentation, you must ensure your project addresses the following:

- **Navigation and Information Architecture.** Logical grouping of menu items and efficient user flow.
- **Feedback and Interaction.** Clear responses to user actions (e.g., hover effects, active states, loading indicators).
- **Control and Error Prevention.** Clear "back" or "home" options and intuitive controls to prevent user frustration.
- **Visual Design and Accessibility.** Contrast and color compatibility, legible typography, and the use of intuitive icons.
- **Affordance and Efficiency:** Use of shortcuts (where applicable) and clear visual cues that suggest how components should be used.

Presentation and Defense Requirements. You must prepare a presentation (5–15 minutes) that includes:

For every major UI element, explain why it was designed that way

Identify at least one UI challenge you faced during development and explain how you solved it to improve the user experience.

Live Demonstration. A walkthrough of the navigation and key interactive components.

Deliverables

- **Format.** Presentation slides (PowerPoint, Canva, etc.) OR a recorded video demonstration with a voice-over.
- **Documentation.** A brief summary (PDF or Word) listing the 5–10 key HCI principles you implemented.
- **Cover Page.** Must include the Author's Name, Project Title, and a link to the project (if hosted) or a GitHub repository.

IMPORTANT ADDITIONS

THE WEBSITE/PROTOTYPE MUST SATISFY THE FOLLOWING:

- Navigation and Structure. Menu depth, logic of element grouping, and search efficiency.

- Interaction and Feedback. How does the system respond to user actions (hovering, clicking, loading indicators)? Is the user's current location within the system clear?
- Clarity of Control and Error Prevention. Can the user easily correct mistakes (e.g., "Undo" action)? Does the system help prevent errors from occurring?
- Visual Consistency. Suitability of the color palette, intuitiveness of icons, legibility of fonts, and the stylistic consistency of components.
- Efficiency. Support for keyboard shortcuts or other methods to speed up the workflow for experienced users.

THE WEBSITE/PROTOTYPE MUST ALSO SCORE PERFECTLY ACCORDING TO **Jakob Nielsen's 10 usability heuristics represent the 'gold standard' within the field of UX/UI design.**

1. Visibility of system status

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.

Example: A progress bar showing how much of a file has been uploaded.

2. Match between system and the real world

The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon.

Example: Using a "trash can" icon for deleting files.

3. User control and freedom

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.

Example: An "Undo" button after accidentally deleting an email.

4. Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions.

Example: Placing the "Settings" icon as a gear shape, as users expect this.

5. Error prevention

Good error messages are important, but the best designs carefully prevent problems from occurring in the first place.

Example: Graying out a "Submit" button until all required fields are filled correctly.

6. Recognition rather than recall

Minimize the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another.

Example: A "Recently Viewed" section in an e-commerce store.

7. Flexibility and efficiency of use

Shortcuts — hidden from novice users — may speed up the interaction for the expert user such that the design can cater to both inexperienced and experienced users.

Example: Keyboard shortcuts (Ctrl+C, Ctrl+V) for power users.

8. Aesthetic and minimalist design

Interfaces should not contain information which is irrelevant or rarely needed. Every extra unit of information competes with the relevant units of information.

Example: A clean Google search homepage vs. a cluttered news portal.

9. Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.

Example: Instead of "Error 404", saying "The page you are looking for does not exist. Try going back to the Home page."

10. Help and documentation

It’s best if the system doesn’t need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

Example: A searchable "Help Center" or tooltips explaining complex features.

WE ARE ALSO TO MAKE SURE IT LACKS THE ERRORS/ISSUES FOUND IN THE WEBSITE ANALYSED IN THE PPT FILE

1. Visibility of system status

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.

Example: A progress bar showing how much of a file has been uploaded.

2. Match between system and the real world

The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon.

Example: Using a "trash can" icon for deleting files.

3. User control and freedom

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.

Example: An "Undo" button after accidentally deleting an email.

4. Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions.

Example: Placing the "Settings" icon as a gear shape, as users expect this.

5. Error prevention

Good error messages are important, but the best designs carefully prevent problems from occurring in the first place.

Example: Graying out a "Submit" button until all required fields are filled correctly.

6. Recognition rather than recall

Minimize the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another.

Example: A "Recently Viewed" section in an e-commerce store.

7. Flexibility and efficiency of use

Shortcuts — hidden from novice users — may speed up the interaction for the expert user such that the design can cater to both inexperienced and experienced users.

Example: Keyboard shortcuts (Ctrl+C, Ctrl+V) for power users.

8. Aesthetic and minimalist design

Interfaces should not contain information which is irrelevant or rarely needed. Every extra unit of information competes with the relevant units of information.

Example: A clean Google search homepage vs. a cluttered news portal.

9. Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.

Example: Instead of "Error 404", saying "The page you are looking for does not exist. Try going back to the Home page."

10. Help and documentation

It’s best if the system doesn’t need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

Example: A searchable "Help Center" or tooltips explaining complex features.

It must be perfect in terms of the following points: • Navigation and Structure. Menu depth, logic of element grouping, and search efficiency.

• Interaction and Feedback. How does the system respond to user actions (hovering, clicking, loading indicators)? Is the user’s current location within the system clear?

• Clarity of Control and Error Prevention. Can the user easily correct mistakes (e.g., "Undo" action)? Does the system help prevent errors from occurring?

• Visual Consistency. Suitability of the color palette, intuitiveness of icons, legibility of fonts, and the stylistic consistency of components.

- Efficiency. Support for keyboard shortcuts or other methods to speed up the workflow for experienced users